

|          |                                                                                                                                                                                     |           |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|          |                                                                                                                                                                                     |           |
| 4(a)     | <u>sum/total</u> momentum before = <u>sum/total</u> momentum after<br><b>or</b><br><u>sum/total</u> momentum (of a system of objects) is constant                                   | <b>M1</b> |
|          | <u>if no (resultant) external force/for a closed system</u>                                                                                                                         | <b>A1</b> |
| 4(b)(i)  | $(3.0 \times 4.0 \times \cos \theta)$ <b>or</b> $(2.5 \times 4.8 \times \cos \theta)$ <b>or</b> $(5.5 \times 3.7)$                                                                  | <b>C1</b> |
|          | $(3.0 \times 4.0 \times \cos \theta) + (2.5 \times 4.8 \times \cos \theta) = (5.5 \times 3.7)$                                                                                      | <b>C1</b> |
|          | $\theta = 32^\circ$                                                                                                                                                                 | <b>A1</b> |
| 4(b)(ii) | (initial $E_K = \frac{1}{2} \times 3.0 \times 4.0^2 + \frac{1}{2} \times 2.5 \times 4.8^2 =$ ) 53 (J)<br><b>or</b><br>(final $E_K = \frac{1}{2} \times 5.5 \times 3.7^2 =$ ) 38 (J) | <b>C1</b> |
|          | values of initial $E_K$ and final $E_K$ both correct <b>and</b> inelastic stated                                                                                                    | <b>A1</b> |